

```
(%i6) string(scsimp(expr, eq1, eq2));
(%o6)                                c^2-8
(%i7) quit();
```

Unlike Octave, Maxima, by default, doesn't evaluate its expressions; it only simplifies them. This means that expressions with integers like $\cos(1)$, $\exp(2)$, $\sqrt{3}$, etc., may remain as they are in the most simplified form, without being evaluated to their respective float numerical values. In such cases, we may force evaluation by passing the option *numer*. Similar evaluation can be achieved for predicates, using *pred*:

```
$ maxima -q
(%i1) cos(1);
(%o1)                                cos(1)
(%i2) cos(1), numer;
(%o2)                                .5403023058681398
(%i3) sqrt(7);
(%o3)                                sqrt(7)
(%i4) sqrt(7), numer;
(%o4)                                2.645751311064591
(%i4) 1 + 2 > 9;
(%o4)                                3 > 9
(%i5) 1 + 2 > 9, pred;
(%o5)                                false
(%i6) string(%pi < %pi%e); /* string to print it on one line */
(%o6)                                %e%pi < %pi%e
(%i7) %e%pi < %pi%e, pred;
(%o7)                                false
(%i8) quit();
```

Summation simplifications

Symbolical representation and manipulations of summations

can be beautifully achieved by using *sum()* and the option *simpsum*. Check out the code execution below:

```
$ maxima -q
(%i1) sum(a * i, i, 1, 5);
(%o1)                                15 a
(%i2) sum(a, i, 1, 5);
(%o2)                                5 a
(%i3) sum(a * i, i, 1, n);
(%o3)                                n
=====
\
a > i
/
=====
i = 1
(%i4) simpsum: true$ /* Enable simplification of summations */
(%i5) string(sum(a * i, i, 1, n)); /*string to print on a line*/
(%o5)                                a*(n^2+n)/2
(%i6) string(sum(i^2 - i, i, 1, n) + sum(j, j, 1, n));
(%o6)                                (2^n^3+3^n^2+n)/6
(%i7) string(factor(sum(i^2 - i, i, 1, n) + sum(j, j, 1, n)));
(%o7)                                n^(n+1)*(2^n+1)/6
(%i8) quit();
```



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